CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 7: Coordinate Geometry

Previous year board questions, topic-wise: 25 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 The distance formula

1. The centre of a circle is $(x-7, 2x)$. The circle passes through $(-9, 11)$ and its radius is $5\sqrt{2}$ units. Find the possible values of x. CBSE 2026, 2 marks
2. A circle has its centre at $(2a, a-7)$ and passes through the point $(11, -9)$. Its diameter is $10\sqrt{2}$ units. Find the values of a. CBSE 2025, 2 marks
3. $\triangle ABC$ has vertices $A(5, -6)$, $B(6, 4)$ and $C(0, 0)$. Find the length of its median AD. (a) $\sqrt{68}$ units (b) $2\sqrt{15}$ units (c) $\sqrt{101}$ units (d) 10 units CBSE 2024, 1 mark
4. The points $(3, -5)$ and $(x, -5)$ are 15 units apart. Find the possible values of x. (a) $12, -18$ (b) $-12, 18$ (c) $18, 5$ (d) $-9, -12$ CBSE 2024, 1 mark
5. Find the distance between $(0, 2\sqrt{5})$ and $(-2\sqrt{5}, 0)$. (a) $2\sqrt{10}$ units (b) $4\sqrt{10}$ units (c) $2\sqrt{20}$ units (d) 0 CBSE 2023, 1 mark
6. Two vertices of an equilateral triangle are $(-5, 3)$ and $(5, 3)$, and the origin lies inside the triangle. Find the third vertex. Take $\sqrt{3} = 1.7$. CBSE 2023, 3 marks
7. Assertion (A): $P(0, 2)$ is the point where the line $3x + 2y = 4$ meets the y-axis. Reason (R): The distance of $P(0, 2)$ from the x-axis is 2 units. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true CBSE 2023, 1 mark
8. Find the point on the y-axis that is equidistant from $(5, -2)$ and $(-3, 2)$. CBSE 2019, 3 marks
9. Write the distance of the point $P(x, y)$ from the origin. CBSE 2018, 1 mark
10. Show that $(3, 0)$, $(6, 4)$ and $(-1, 3)$ are the vertices of a right-angled isosceles triangle. CBSE 2016, 2 marks
11. The point $P(x, y)$ is equidistant from $A(a+b, b-a)$ and $B(a-b, a+b)$. Prove that $bx = ay$. CBSE 2016, 3 marks

2 The section formula

12. Where does the mid-point of the segment joining $(5, -4)$ and $(6, 4)$ lie? (a) on the x-axis (b) on the y-axis (c) at the origin (d) on neither axis CBSE 2026, 1 mark
13. In what ratio does the x-axis divide the segment joining $(-6, 5)$ and $(-4, -1)$? Find the point where they meet. CBSE 2026, 3 marks
14. Where does the mid-point of the segment joining $P(-4, 5)$ and $Q(4, 6)$ lie? CBSE 2025, 1 mark

15. In what ratio does the y-axis divide the segment joining $(5, -6)$ and $(-1, -4)$? Find the point where they meet. CBSE 2025, 3 marks

16. A circle has centre $(2, 0)$, and one end of a diameter is $(6, 0)$. Find the other end. (a) $(0, 0)$ (b) $(4, 0)$ (c) $(-2, 0)$ (d) $(-6, 0)$ CBSE 2024, 1 mark

17. The point $\left(\frac{8}{5}, y\right)$ lies on the segment joining $(1, 2)$ and $(2, 3)$. In what ratio does it divide the segment? Find y as well. CBSE 2024, 3 marks

18. $A(-1, -1)$, $B(-1, 6)$, $C(3, 6)$ and $D(3, -1)$ are the vertices of a rectangle. P, Q, R and S are the mid-points of AB, BC, CD and DA. Show that the diagonals of $PQRS$ bisect each other. CBSE 2024, 3 marks

19. In what ratio does the x-axis divide the segment joining $A(3, 6)$ and $B(-12, -3)$? (a) $1 : 2$ (b) $1 : 4$ (c) $4 : 1$ (d) $2 : 1$ CBSE 2023, 1 mark

20. AB is a diameter of a circle with centre $(2, -3)$, and B is the point $(1, 4)$. Find the coordinates of A. CBSE 2019, 1 mark

21. In what ratio does the x-axis divide the segment joining $(1, -3)$ and $(4, 5)$? Find the point of division. CBSE 2019, 2 marks

22. The segment joining $A(2, 1)$ and $B(5, -8)$ is trisected at P and Q, with P nearer to A. P also lies on the line $2x - y + k = 0$. Find k. CBSE 2019, 3 marks

23. In what ratio does $P(4, m)$ divide the segment joining $A(2, 3)$ and $B(6, -3)$? Find m. CBSE 2018, 2 marks

24. $A(-2, 1)$, $B(a, 0)$, $C(4, b)$ and $D(1, 2)$ are the vertices of a parallelogram $ABCD$. Find a and b, and the lengths of its sides. CBSE 2018, 3 marks

25. P and Q are the points of trisection of the segment joining $A(2, -2)$ and $B(-7, 4)$, with P nearer to A. Find P and Q. CBSE 2016, 2 marks

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